

Physics-guided discriminability of process-level observables in partially observable cancer systems

Sin Guan Kong^{1, 2, *}

¹Department of Electrical and Electronic Engineering, Lee Kong Chian Faculty of Engineering and Science, Universiti Tunku Abdul Rahman, Bandar Sungai Long, 43000 Kajang, Selangor, Malaysia

²Centre for Healthcare Science and Technology, Universiti Tunku Abdul Rahman, Bandar Sungai Long, 43000 Kajang, Selangor, Malaysia

*Corresponding author: kongsg@utar.edu.my

Abstract

Bulk gene expression measurements are widely used to construct pathway- and process-level representations of cancer systems. However, such measurements are not direct readouts of unique biological states, but effective observations shaped by cellular mixtures, biological constraints, and measurement uncertainty. Under partial observability, similar observable patterns may therefore correspond to heterogeneous underlying system organizations, limiting direct interpretation of gene-expression-derived process observables.

In this study, we propose a physics-guided framework for interpreting process-level gene expression observables through the concepts of observable state clouds, cloud thickness, and discriminability. Rather than representing biological processes solely by scalar activity scores, each biological process (BP) observable is characterized using both its mean activity (μ) and its full width at half maximum (FWHM, W). Within this framework, W serves as an operational proxy for observable uncertainty and cloud thickness, reflecting the combined effects of biological fluctuation, projection ambiguity, and measurement variability.

Using multiple cancer gene expression cohorts, we systematically analyzed the geometric organization of Hallmark and BP-level observables across cancer samples. The analyses revealed substantial variation in cloud thickness and coherence among biological processes, even under comparable activity levels.

Structured biological process observables further exhibited organization patterns distinct from size-matched random gene sets, supporting the role of biological constraints in shaping observable geometry. Based on these observations, we define a physics-guided discriminability formulation integrating signal magnitude and cloud thickness for evaluating process-level observables under projection uncertainty.

These results suggest that process-level gene expression observables should not be interpreted solely by activity magnitude, but also by their associated uncertainty structure and observable geometry. More broadly, this work provides a measurement-aware perspective for interpreting biological process observables in partially observable cancer systems.

Keywords:

Partial observability; Gene expression; Biological processes; Hallmark pathways; Discriminability; Observable geometry; Uncertainty structure; Systems biology

1. Introduction

Cancer systems are commonly studied through molecular observables derived from bulk gene expression measurements, including pathway scores, biological process activities, and transcriptomic signatures. These observables are widely used for cancer classification, survival stratification, and mechanistic interpretation because they provide compact representations of high-dimensional molecular measurements. In particular, Hallmark and biological process (BP) gene sets have become important tools for interpreting collective biological activity at the systems level [1–6].

However, gene-expression-derived observables are not direct measurements of unique biological states. Bulk gene expression profiles arise from heterogeneous mixtures of tumor cells, immune cells, stromal components, vascular structures,

and other microenvironmental elements, together with biological fluctuation and measurement variability [10–14]. As a result, similar observable patterns may correspond to heterogeneous underlying biological organizations. This limitation becomes particularly important in cancer systems, where tumor heterogeneity, clonal evolution, and microenvironmental diversity strongly influence observable molecular patterns and therapeutic responses [13–17].

Existing pathway-level analyses, including gene set enrichment approaches and pathway activity scoring methods, primarily represent biological processes using scalar activity values derived from grouped gene expression measurements [5–8]. While such approaches are highly useful for dimensionality reduction and biological interpretation, they generally focus on activity magnitude rather than the uncertainty structure of the observable itself. In practice, two biological processes may exhibit comparable activity levels while displaying substantially different degrees of internal dispersion, coherence, or fluctuation.

From a systems perspective, this limitation reflects a broader problem of partial observability. Observable molecular measurements can be regarded as projected representations of underlying biological organization rather than direct readouts of latent system states [22–26]. Under non-injective observation mappings, multiple latent configurations may generate similar observable patterns, making inverse reconstruction intrinsically underdetermined. Consequently, observable similarity does not necessarily imply similarity of underlying biological organization or clinical behavior.

In physics and complex systems theory, collective observables are often characterized not only by their mean behavior but also by their fluctuation structure [27–31]. Macroscopic observables such as density, magnetization, or order parameters are commonly interpreted together with their associated variance or fluctuation properties. Motivated by this perspective, we propose that process-level gene expression observables should similarly be interpreted as observable state clouds rather than scalar activity values alone. Within this

framework, the mean activity of a biological process represents the cloud center, whereas the full width at half maximum (FWHM) of the observable distribution represents cloud thickness and serves as an operational proxy for observable uncertainty.

Here, we present a physics-guided framework for interpreting process-level gene expression observables in partially observable cancer systems. Instead of treating biological process observables solely as pathway activity scores, we characterize each process-level observable using both its mean activity (μ_B) and cloud thickness (W). Using Hallmark and BP-level observables across multiple cancer cohorts, we investigate how observable geometry, cloud thickness, and uncertainty structure shape process-level discriminability. More broadly, this work aims to provide a measurement-aware perspective for interpreting gene-expression-derived process observables in cancer systems.

2. Process-level observables as projected state clouds

Bulk gene expression measurements represent effective observations of complex biological systems rather than direct measurements of unique latent states. In cancer tissues, observed gene expression signals arise from heterogeneous mixtures of tumor cells, immune populations, stromal components, vascular structures, metabolic conditions, and signaling interactions [10–17,34,35]. Consequently, gene expression observables should be interpreted as projected representations shaped by both biological organization and measurement processes.

Within this perspective, the observed expression value of an individual gene may be represented as:

$$O_g^{obs} = h_g(S; C_{bio}) + \epsilon_g$$

where S denotes the underlying latent biological organization, h_g represents the observation mapping, C_{bio} denotes biological constraints and contextual

interactions, and ϵ_g represents measurement variability and stochastic fluctuation. Under partial observability, the mapping h_g is generally non-injective, implying that multiple latent biological configurations may generate similar observable values [22–26].

Importantly, uncertainty in gene expression observables does not arise solely from technical noise. Observable ambiguity may emerge from several interacting sources, including biological heterogeneity, projection effects induced by mixed cellular composition, pathway redundancy, compensatory signaling behavior, and measurement variability [13,14,35–38]. As a result, observable similarity does not necessarily imply similarity of underlying biological organization.

Although single-gene measurements may contain latent uncertainty structure, such structures are typically inaccessible from isolated scalar observables alone. In contrast, biological process (BP) and Hallmark gene sets provide collective observable ensembles that enable operational characterization of process-level organization [1–6]. Rather than treating pathway observables solely as averaged activity scores, we interpret them as projected state clouds formed by coordinated gene-expression-derived observables.

To operationalize uncertainty-bearing process-level observables under partial observability, we constructed biological-process (BP) observables in a joint activity–uncertainty space defined by BP activity magnitude and dispersion structure. Within this representation, each BP observable is characterized by both its mean activity (μ) and observable cloud thickness (W), enabling simultaneous interpretation of collective activity and uncertainty structure.

Figure 1. Conceptual framework of uncertainty-aware discriminability in transcriptomic observables

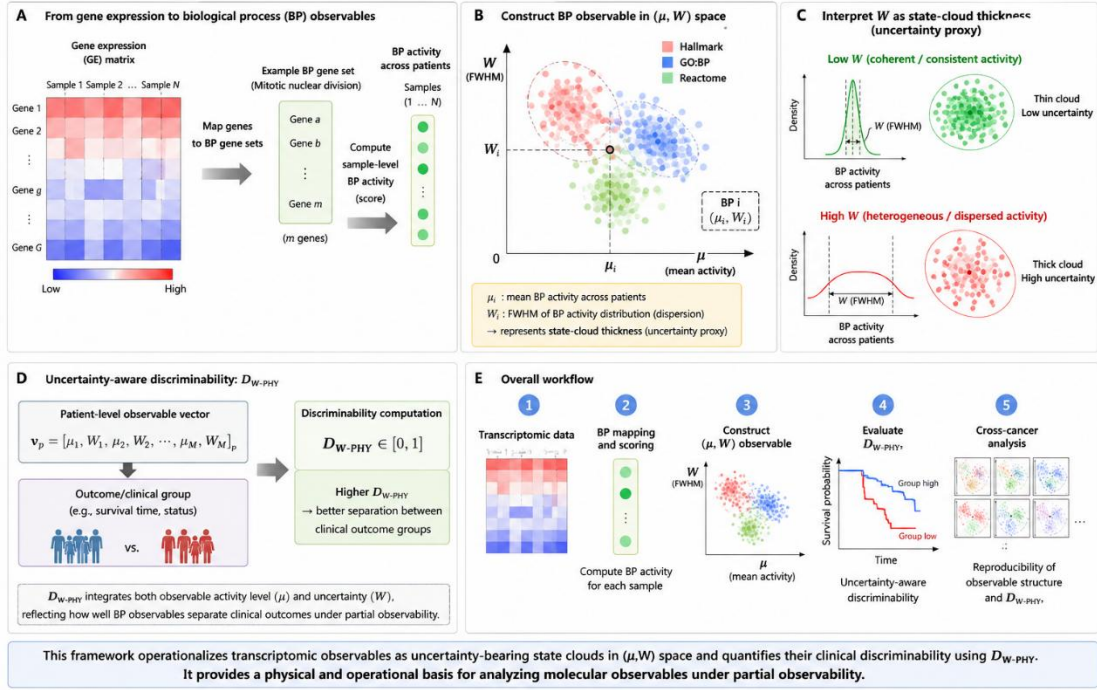


Figure 1. Conceptual framework of uncertainty-aware discriminability in process-level gene expression observables

(A) Bulk gene expression (GE) profiles were mapped onto biological process (BP) gene sets to construct sample-level BP observables. Each BP observable represents a collective process-level measurement derived from grouped gene expression signals.

(B) BP observables were represented in a joint (μ, W) space, where μ denotes the mean BP activity across samples and W denotes the full width at half maximum (FWHM) of the BP activity distribution. This representation enables simultaneous characterization of observable magnitude and dispersion structure.

(C) Observable cloud thickness (W) was interpreted as an operational proxy for uncertainty structure under partial observability. Low- W observables form coherent and compact state clouds, whereas high- W observables exhibit broader and more heterogeneous observable organization.

(D) Patient-level observable vectors integrating BP activity and uncertainty structure were used to compute uncertainty-aware discriminability (D_{W-PHY}), reflecting the ability of process-level observables to separate clinical outcome groups under partial observability.

(E) Overall analytical workflow, including transcriptomic preprocessing, BP mapping, construction of BP observables in (μ, W) space, uncertainty-aware discriminability evaluation, and cross-cancer analysis.

Together, this framework operationalizes process-level gene expression observables as uncertainty-bearing observable state clouds and provides a physics-guided perspective for analyzing molecular observables under partial observability.

For a biological process B containing N_B genes, the process-level observable is represented as:

$$O_B = \{Z_{g_1}, Z_{g_2}, \dots, Z_{g_{N_B}}\}$$

where Z_g denotes the normalized expression value of gene g . Within this framework, each biological process is characterized by two complementary quantities:

$$\mu_B = \frac{1}{N_B} \sum_{g \in B} Z_g$$

and

$$W_B = \text{FWHM}(B)$$

where μ_B represents the mean process activity and W_B represents the full width at half maximum (FWHM) of the process-level expression distribution. Here, μ_B is interpreted as the center of the observable state cloud, whereas W_B represents cloud thickness and serves as an operational proxy for observable uncertainty.

Importantly, W_B should not be interpreted solely as technical noise. Instead, observable cloud thickness reflects the combined effects of biological fluctuation, projection ambiguity, collective coordination looseness, and measurement variability. Biological processes exhibiting similar mean activity may therefore display substantially different uncertainty structures and coherence properties.

Under this framework, biological process observables are not interpreted as

direct representations of unique latent biological states, but as coarse-grained collective projections of partially observable system organization [30–33,47–49]. The resulting observable geometry provides an operational representation of process-level organization under projection uncertainty.

3. Physics-guided discriminability of process-level observables

Conventional pathway-level analyses primarily evaluate biological processes using activity magnitude alone. However, under partial observability, observable activity does not fully characterize the reliability or coherence of a process-level representation. Biological processes exhibiting similar mean activity may possess substantially different internal dispersion structures, reflecting differences in collective organization, projection ambiguity, or fluctuation behavior.

Within the proposed framework, process-level observables are interpreted as observable state clouds characterized by both cloud center and cloud thickness. Accordingly, biological process observables are represented using the pair:

$$(\mu_B, W_B)$$

where μ_B denotes mean process activity and W_B denotes observable cloud thickness measured using the full width at half maximum (FWHM). This representation enables simultaneous characterization of activity magnitude and uncertainty structure.

Under this interpretation, several qualitatively distinct observable regimes emerge. Biological processes with high μ_B and low W_B represent coherently activated observables with relatively constrained internal organization. In contrast, processes with high μ_B but large W_B indicate broader observable dispersion and reduced coherence despite comparable activity levels. Similarly, low-activity processes may exhibit either stable suppression or highly dispersed organization depending on their cloud thickness. Thus, observable cloud geometry provides additional information beyond scalar activity scores alone.

To operationalize this concept, we define a physics-guided discriminability framework integrating both observable magnitude and cloud thickness. For a system containing K process-level observables, we define:

$$M = \frac{1}{K} \sum_{B=1}^K |\mu_B|$$

and

$$U = \frac{1}{K} \sum_{B=1}^K \log(W_B + \delta)$$

where M represents average observable magnitude, U represents average cloud-thickness-associated uncertainty, and δ is a small constant preventing numerical divergence. Based on these quantities, the system-level physics-guided discriminability is defined as:

$$D_{W-PHY} = M - \lambda U$$

where λ controls the contribution of cloud thickness to the overall discriminability formulation.

Within this framework, high discriminability emerges not solely from strong observable magnitude, but from the balance between signal strength and observable coherence. Biological processes with large activity magnitude but broad cloud thickness may possess reduced interpretability due to increased observable uncertainty. Conversely, processes with relatively constrained cloud structure may provide more coherent representations of collective biological organization.

To evaluate the operational behavior of uncertainty-aware discriminability under different uncertainty-weighting conditions, we systematically analyzed the influence of cloud-thickness penalties across multiple cancer cohorts.

A

Figure 4. Real-data uncertainty sensitivity of process-level discriminability

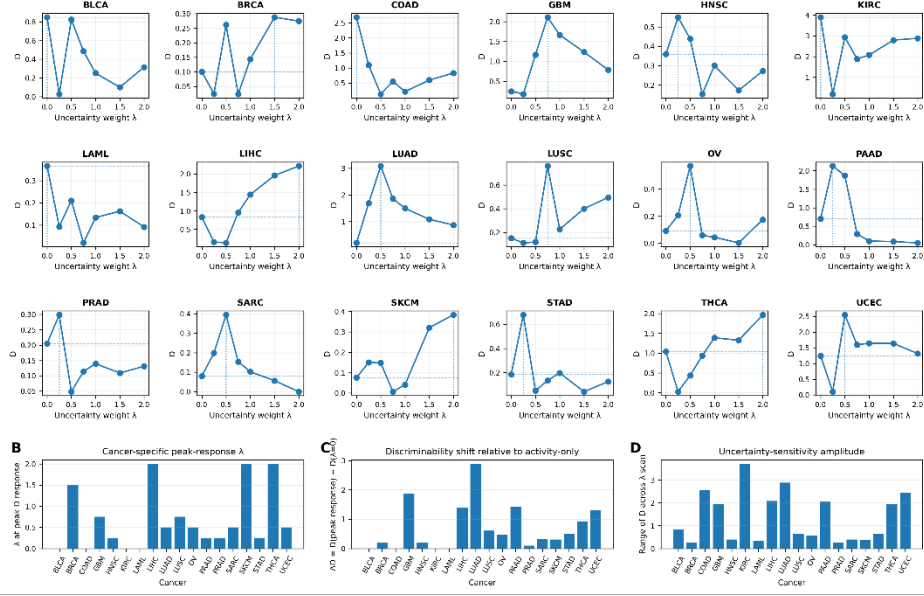


Figure 2. Process-level discriminability exhibits cancer-specific responses to uncertainty weighting

(A) Cancer-specific responses of uncertainty-aware discriminability (D) across different uncertainty-weighting conditions (λ). Different cancer cohorts exhibited distinct discriminability response patterns as the contribution of observable cloud thickness was varied. Dashed horizontal lines indicate the activity-only baseline ($\lambda = 0$), while vertical dashed lines indicate the peak-response uncertainty-weighting region for each cancer.

(B) Summary of cancer-specific uncertainty-response regimes. Different cancer types exhibited heterogeneous peak-response weighting regions, suggesting that the influence of observable cloud thickness on discriminability varies across partially observable cancer systems.

(C) Discriminability shifts relative to activity-only observables ($\lambda = 0$). Incorporating uncertainty structure altered process-level discriminability behavior across cancers, with varying degrees of sensitivity to cloud-thickness weighting.

(D) Uncertainty-sensitivity amplitude across cancers, measured as the range of discriminability values observed across uncertainty-weighting conditions. Certain cancers exhibited substantially larger uncertainty-response variability,

indicating stronger dependence of discriminability on observable cloud-thickness organization.

Together, these results demonstrate that process-level discriminability responds non-uniformly to uncertainty weighting across cancers, supporting the contribution of cloud-thickness organization beyond activity magnitude alone under partial observability.

Importantly, the proposed framework does not attempt to reconstruct unique latent biological states from observable measurements. Instead, it provides an operational approach for evaluating observable geometry under partial observability. In this context, cloud thickness represents a measurable trace of uncertainty structure rather than a direct estimate of latent entropy itself [27–30].

From a systems perspective, the proposed formulation is conceptually related to fluctuation-aware interpretations commonly used in statistical physics and complex systems analysis [29–33,47–49]. In such systems, collective observables are often characterized not only by average behavior, but also by their fluctuation structure and coherence properties. Here, process-level gene expression observables are similarly interpreted as coarse-grained collective observables whose geometry reflects partially observable biological organization.

Together, this framework establishes a measurement-aware interpretation of process-level gene expression observables in cancer systems, where discriminability depends jointly on observable magnitude and cloud thickness rather than activity level alone.

4. Observable geometry of biological processes across cancer systems

To investigate the organization of process-level observables under partial observability, we analyzed Hallmark and biological process (BP) gene sets across

multiple cancer gene expression cohorts obtained from publicly available cancer transcriptome datasets [16–18]. For each biological process, the observable state cloud was characterized using its mean activity (μ_B) and cloud thickness (W_B), enabling systematic examination of process-level observable geometry across cancer systems.

The resulting BP observable landscapes revealed substantial heterogeneity in uncertainty structure and coherence across cancers and gene-set databases.

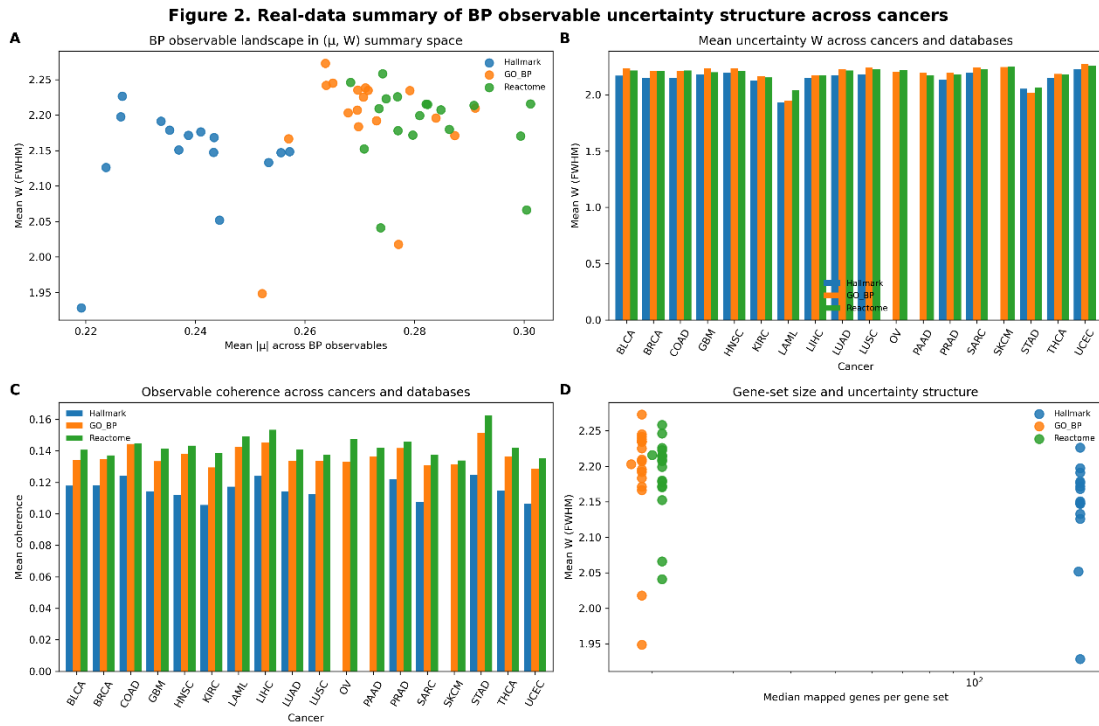


Figure 3. Real-data organization of BP observable uncertainty structure across cancers

(A) Global BP observable landscape in joint (μ, W) space across multiple cancer cohorts and gene-set databases. Each point represents the average observable geometry of a cancer-specific BP collection, where μ denotes mean BP activity magnitude and W denotes observable cloud thickness measured using full width at half maximum (FWHM). Distinct distributions emerge across Hallmark, GO:BP, and Reactome observables.

(B) Mean observable cloud thickness (W) across cancers and gene-set databases.

Systematic differences in uncertainty structure were observed across cancer types and BP collections, indicating heterogeneous organization of process-level observable dispersion.

(C) Mean observable coherence across cancers and databases. Coherence was quantified using the inverse relationship between observable magnitude and cloud thickness, reflecting the compactness of BP observable organization under partial observability.

(D) Relationship between gene-set size and observable cloud thickness. Different BP databases occupy distinct regions in gene-set-size versus uncertainty space, suggesting that observable geometry is influenced by both biological organization and collective feature scale.

Together, these results demonstrate that process-level gene expression observables exhibit structured and heterogeneous uncertainty organization across cancers rather than uniform dispersion behavior.

Global analyses revealed substantial heterogeneity in observable cloud structure among biological processes. While some Hallmark observables displayed relatively constrained cloud thickness with coherent activity organization, others exhibited broad observable dispersion despite comparable activity magnitude. These observations indicate that process-level activity alone is insufficient to characterize observable organization. Instead, biological processes possessing similar mean activity may correspond to markedly different uncertainty structures.

Across cancer cohorts, coherent process-level observables frequently appeared as regions combining relatively elevated μ_B with constrained W_B . In contrast, broader cloud thickness was often associated with increased observable heterogeneity and reduced process coherence. Importantly, cloud-thickness variation was not uniformly distributed across biological processes, suggesting that observable geometry is shaped by underlying biological constraints rather than random fluctuation alone.

To further evaluate whether the observed geometry reflected biologically structured organization, we compared Hallmark and BP observables against size-matched random gene sets [3–6]. Although certain random gene sets occasionally exhibited elevated discriminability, structured biological process observables displayed distinct organization patterns in terms of cloud thickness distribution, coherence behavior, and observable geometry. These results suggest that biological constraints contribute to shaping process-level observable structure beyond simple aggregation effects.

Structured biological process observables exhibited distinct uncertainty organization patterns compared with random gene-set controls across multiple cancers.

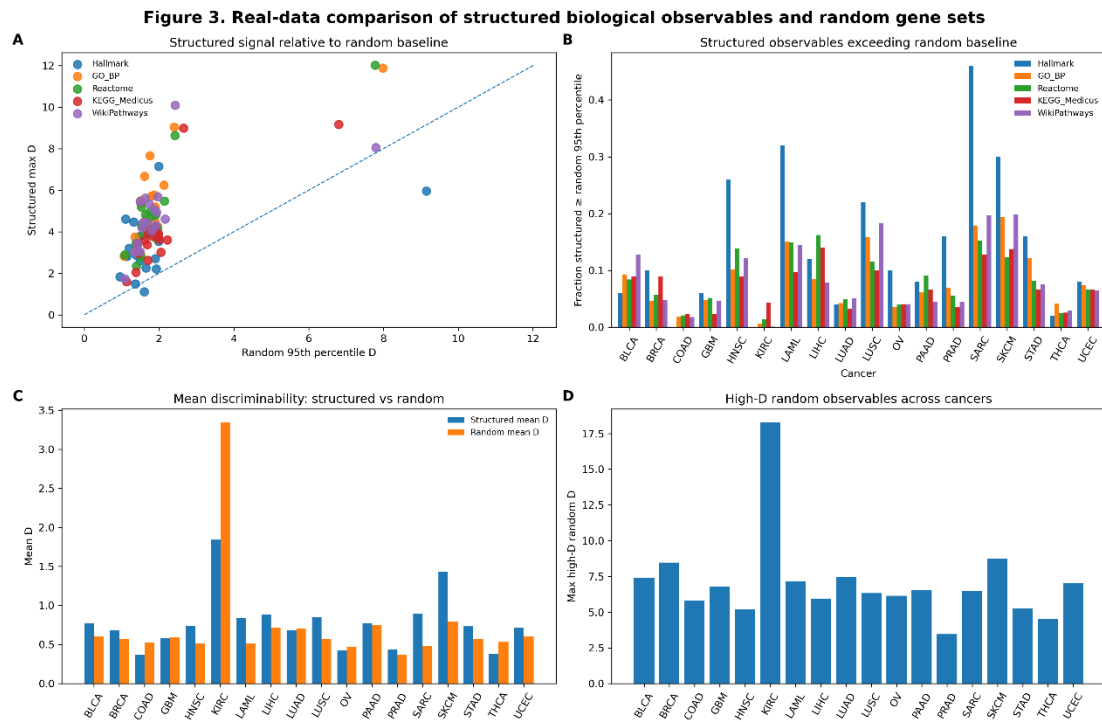


Figure 4. Structured biological-process observables exhibit non-random uncertainty organization

(A) Representative distributions of uncertainty-aware discriminability (D_{W-PHY}) for structured Hallmark observables and size-matched random gene sets across representative cancer cohorts. Structured BP observables displayed distinct

discriminability organization patterns relative to random controls.

(B) Comparison of observable cloud thickness (W) distributions between Hallmark observables and random gene sets. Structured BP observables exhibited characteristic uncertainty organization distinct from randomly aggregated gene collections.

(C) Coherence structure of structured and random observables in joint (μ, W) space. Hallmark observables formed organized regions with constrained uncertainty structure, whereas random gene sets exhibited broader and less structured dispersion behavior.

(D) Cross-cancer summary of structured-versus-random comparisons. Although certain random gene sets occasionally achieved elevated discriminability, structured biological process observables consistently displayed distinct uncertainty organization and coherence behavior across cancers.

Together, these results support the interpretation that observable geometry and cloud-thickness organization are shaped, at least partially, by biologically constrained collective structure rather than by random gene aggregation alone.

Notably, observable cloud thickness should not be interpreted solely as technical variability. Under the proposed framework, W_B reflects a composite effect arising from biological fluctuation, projection ambiguity, collective coordination looseness, and measurement variability. Consequently, broader cloud thickness may indicate reduced observable coherence even when process activity remains elevated. This interpretation is consistent with the view that bulk gene expression measurements represent projected observations of heterogeneous biological organization rather than direct measurements of unique latent states [10–17,34–38].

From a systems perspective, Hallmark and BP observables may therefore be interpreted as coarse-grained collective observables shaped by biological constraints and partial observability. In this context, observable geometry provides an operational representation of process-level organization under

projection uncertainty. Rather than evaluating biological processes solely through scalar activity scores, the proposed framework incorporates both observable magnitude and uncertainty structure into process-level interpretation.

Observable geometry and uncertainty structure varied substantially across biological-process collections, suggesting that process-level organization depends not only on activity magnitude but also on the scale and composition of collective observables. To further investigate the influence of collective observation scale under partial observability, we systematically evaluated uncertainty-aware discriminability across different numbers of BP observables.

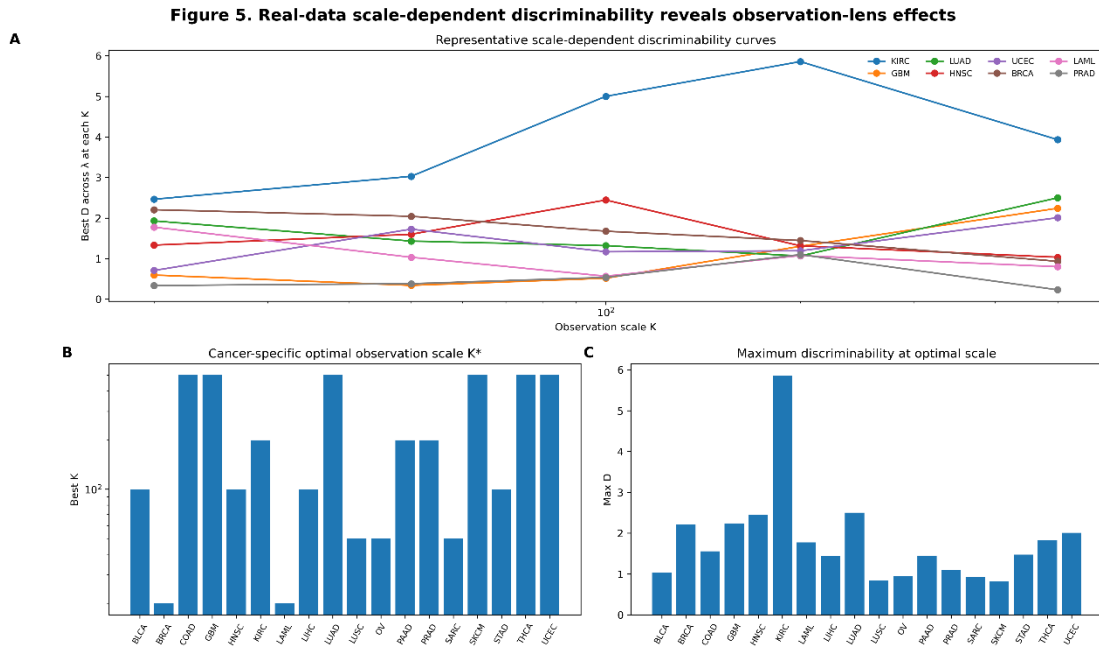


Figure 5. Scale dependence of uncertainty-aware discriminability across collective BP observables

(A) Representative examples of uncertainty-aware discriminability (D_{W-PHY}) evaluated across different numbers of biological-process observables (K) in selected cancer cohorts. Distinct cancers exhibited heterogeneous discriminability behavior as collective observation scale increased.

(B) Cross-cancer comparison of discriminability trajectories under increasing BP observation scales. Observable discriminability frequently displayed non-monotonic behavior rather than uniform improvement with increasing feature

number.

(C) Relationship between collective BP observation scale and observable cloud organization. Increasing numbers of BP observables altered the balance between activity magnitude and cloud-thickness-associated uncertainty across cancers.

(D) Summary distribution of uncertainty-aware discriminability across cancers and observational scales. The results suggest that process-level discriminability depends on collective observation scale under partial observability rather than solely on increasing observable quantity.

Together, these findings support the interpretation that uncertainty-aware discriminability exhibits scale-dependent behavior in process-level gene expression observables, consistent with collective organization effects in partially observable cancer systems.

Importantly, the present framework does not claim direct reconstruction of latent cancer states. Instead, it provides an operational approach for examining how observable geometry and cloud thickness influence process-level discriminability under partially observable conditions. The resulting representation establishes a measurement-aware interpretation of biological process observables in cancer systems.

5. Discussion

In this study, we proposed a physics-guided framework for interpreting process-level gene expression observables in partially observable cancer systems. Rather than representing biological processes solely through scalar activity measurements, the proposed framework interprets Hallmark and BP-level observables as projected state clouds characterized by both observable magnitude and cloud thickness. Within this formulation, the mean activity (μ_B) represents the center of the observable cloud, whereas the full width at half maximum (W_B) provides an operational proxy for observable uncertainty and

cloud thickness.

The proposed framework differs from conventional pathway-level analyses in an important conceptual aspect. Existing pathway scoring approaches primarily quantify activity magnitude [5–8], whereas the present framework additionally considers the geometric and uncertainty structure of the observable itself. Under partial observability, biological processes exhibiting similar activity levels may nevertheless possess substantially different internal dispersion structures. Consequently, observable activity alone may provide an incomplete representation of process-level organization.

Importantly, the proposed framework does not interpret biological process observables as direct representations of unique latent biological states. Instead, BP observables are interpreted as coarse-grained collective projections of partially observable biological organization [22–26,30–33]. In this context, cloud thickness reflects a composite effect arising from biological fluctuation, projection ambiguity, collective coordination looseness, and measurement variability. Observable geometry therefore provides an operational representation of uncertainty structure rather than a direct measurement of latent entropy itself. The comparison between structured biological process observables and size-matched random gene sets further suggests that biological constraints contribute to shaping observable geometry. Although random gene sets may occasionally exhibit elevated discriminability, structured Hallmark and BP observables displayed distinct cloud-thickness organization and coherence behavior. These findings support the interpretation that process-level observable geometry is influenced by biologically constrained collective organization rather than simple aggregation effects alone.

From a systems perspective, the proposed framework is conceptually related to fluctuation-aware interpretations commonly used in statistical physics and complex systems analysis [27–33,47–49]. Macroscopic observables in complex systems are often interpreted together with their fluctuation structure rather than through average behavior alone. Similarly, process-level gene expression observables may contain interpretable uncertainty structure beyond scalar activity magnitude. Within this view, biological processes are not treated as

isolated pathway scores, but as collective observables shaped by partial observability and projection constraints.

Several limitations should be noted. First, the present study focuses primarily on bulk gene-expression-derived process observables and does not directly incorporate spatial, temporal, or single-cell resolution measurements. Second, observable cloud thickness represents an operational uncertainty proxy rather than a direct decomposition of biological, projection, and measurement-specific variability components. Third, the current framework treats process-level observables using simplified collective representations and does not yet explicitly model network topology, regulatory hierarchy, or cross-process interaction structure.

Nevertheless, the proposed framework establishes a measurement-aware interpretation of biological process observables under partial observability. More broadly, these results suggest that process-level gene expression observables should be interpreted not only through activity magnitude, but also through their associated uncertainty structure and observable geometry.

6. Conclusion

Bulk gene expression measurements provide projected observations of heterogeneous cancer systems rather than direct readouts of unique latent biological states. Under partial observability, biological process observables may therefore contain substantial uncertainty structure arising from biological heterogeneity, projection ambiguity, collective fluctuation, and measurement variability [10–17,22–26].

In this study, we proposed a physics-guided framework for interpreting process-level gene expression observables using the concepts of observable state clouds, cloud thickness, and discriminability. Instead of representing biological processes solely through scalar activity scores, each process-level observable was characterized using both its mean activity (μ_B) and cloud thickness (W_B). Within this framework, W_B serves as an operational proxy for observable

uncertainty and coherence structure.

Systematic analyses across multiple cancer cohorts revealed substantial heterogeneity in observable geometry among Hallmark and biological process observables. Biological processes exhibiting similar activity magnitude frequently displayed markedly different cloud-thickness organization, suggesting that process-level activity alone incompletely captures observable structure. Comparisons with size-matched random gene sets further supported the role of biological constraints in shaping observable geometry and process-level coherence.

Importantly, the proposed framework does not attempt direct reconstruction of latent cancer states. Rather, it provides a measurement-aware interpretation of process-level observables under partial observability. In this context, biological process observables are interpreted as coarse-grained collective projections whose geometry reflects underlying uncertainty structure.

Overall, this work establishes an operational framework linking process-level activity, cloud thickness, and discriminability in cancer gene expression systems. More broadly, these findings suggest that biological process observables should be interpreted not only by their activity magnitude, but also by their associated uncertainty structure and observable geometry [27–33,47–49].

Author Contributions

S.G.K. conceived the study, developed the conceptual framework, performed the analysis, wrote the manuscript, and prepared all figures.

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Conflict of Interest

The author declares no competing interests.

Data Availability

The datasets analyzed in this study are publicly available from The Cancer Genome Atlas (TCGA) via the UCSC Xena platform (<https://xenabrowser.net>).

Code Availability

Code used for data preprocessing, pathway-level scoring, discriminability calculation, and survival analysis has been provided as supplementary code files with this submission.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, generative AI tools were used to assist with language refinement and editing. All content was critically reviewed, validated, and revised by the author. The author takes full responsibility for the accuracy, integrity, and originality of the work.

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